

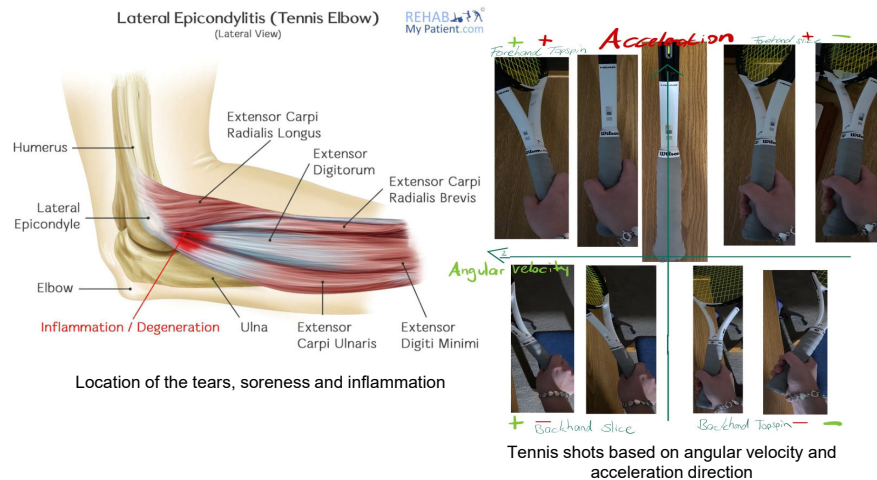
Forecasting Tennis Elbow Using Pressure Sensors

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Background

- Lateral epicondylitis, microscopic tears in the tendon connecting the bony part of the elbow to the muscle in the forearm, is diagnosed by about 50% of all tennis players.¹
- Mainly caused by repetitive arm movement
- Tennis elbow is reduces grip strength



- Prevention techniques commonly include; a good warmup, gentle stretch before playing and using lightweight racquets with bigger grip sizes.²
- The gold standard for testing tennis elbow is to apply gentle pressure to examine your lateral epicondyle and the area above it. In some cases, it will be officially diagnosed using an MRI.
- Topspin, slice, or flat shot can be differentiated by angular velocity and acceleration directions.¹

Objective

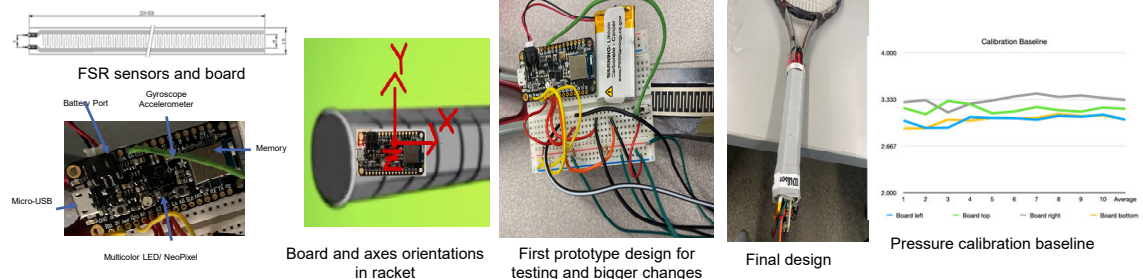
- The objective of this design project is to integrate FSR force sensors and a microcontroller board with inertial sensors. Arduino software collects the pressure and motion data to analyze swing patterns, speed and grip pressure. The device will be tested by healthy and injured players to make predictions on the loss of grip pressure connected to lateral epicondylitis.

Previous Design and Function Ideas



Development Prototype -> Final Version

- Device functions:** attach sensors, turn the device on, sensor calibration of maximum force, start playing, shot detection with angular velocity threshold, measure racket speed, collect force data and compare it to calibration measurement, and red light when danger of injury.
- Adjustments prototype to the final design:** LED strip, smaller design to fit into the grip, solder wires to sensors and board.



Future improvements

- Bluetooth connection and mobile app
- Smaller board size and weight
- Vibration capturing damper
- Detecting volleys and serves

References: 1) Bhakta, Mayuri. Medcor Health Navigation. 2020

2) Skublewska-Paszowska et al.. Sensor Technology for Sports Science. 2020: 3) Adafruit. Adafruit. 2022: